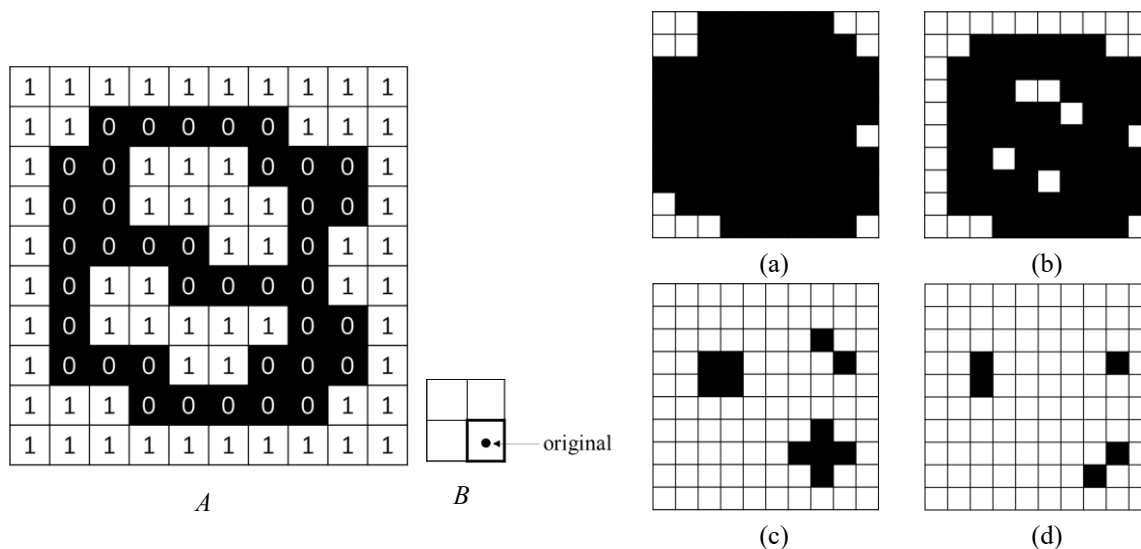


Lab 08

Image Processing II

A. Multiple Choice (10 points)

1. Given an image A and a structuring element B as follow, which of the following image is the result of A eroded by B ($A \ominus B$)?



B. Filters and Convolution (25 points, 5 points for each question)

Generate the following filters and observe the effect on an image. Describe in the report how these filters effect an image or if the filters are extracting any kind of features. (We mentioned some of them in the class)

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(a)	(b)	(c)	(d)	(e)																																													

C. Denoising a Picture (20 points)

Write a program to denoise the image “Saturn.jpg”. The program should reduce the noise level of the image by using different filters. Describe your experiment and include the result images in your report. The filters that you may want to try includes 1) mean filter, 2) gaussian filter, 3) total variation filter, 4) bilateral filter and 5) wavelet denoising filter. Apply the filter to the image for several times and examine if this helps improving the image quality. You may consider to sharpen the image by applying the Laplacian filter to the denoised image.

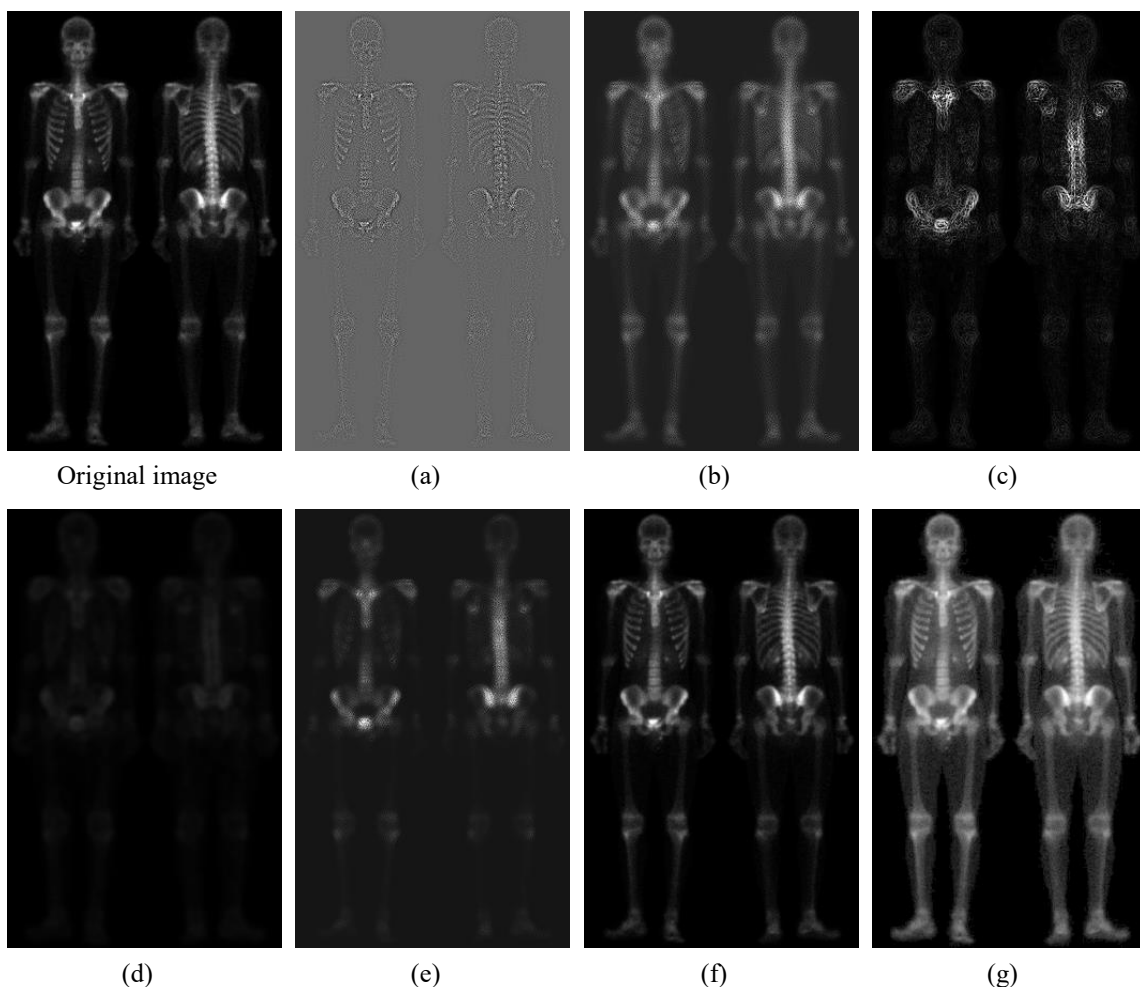
D. Image Enhancement (15 points)

Successful image enhancement is typically not achieved using a single operation. Rather a range of techniques are combined in order to achieve a final result. This example will focus on enhancing an image of bone scan. Apply the following techniques to “Xray.png”:

- (a) Apply Laplacian filter to the original image
- (b) Subtract the image from (a) from the original image to obtain a sharpened image
- (c) Apply both the horizontal and vertical Sobel filter to the image from (b)
- (d) Smooth image from (c) using a 5×5 averaging filter
- (e) Element-by-element multiply the images from (b) and (d)
- (f) Add the image from (e) to the original image to obtain a sharpened image
- (g) Apply power-law transformation (i.e., $g_o = (g_i)^{0.5}$) to adjust the image contrast

Power-law transformation (a.k.a. gamma correction) is an approach to adjust the contrast of an image. In the transformation, the gray level of the pixels was adjusted by the equation $g_o = c(g_i)^r$, here g_o is the gray level of the output image, g_i is the gray level of the input image, and c and r are pre-determined constants. The figure below explains the input-output relationship of power-law transformation. Note that in this example, we set $c = 1$.

The images (pixel values adjusted for displaying) of each step are shown below.



E. Fingerprint Analysis (30 points)

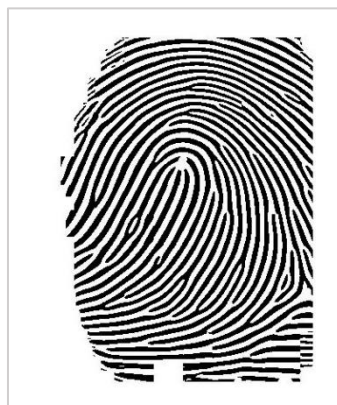
Fingerprint identification is one of the most well-known and publicized biometrics. Because of their uniqueness and consistency over time, fingerprints have been used for identification for over a century, more recently becoming automated (i.e. a biometric) due to advancements in computing capabilities.

A fingerprint usually appears as a series of dark lines that represent the high, peaking portion of the friction ridge skin, while the valley between these ridges appears as white space and are the low, shallow portion of the friction ridge skin. Fingerprint identification is based primarily on the minutiae, or the location and direction of the ridge endings and bifurcations (splits) along a ridge path.

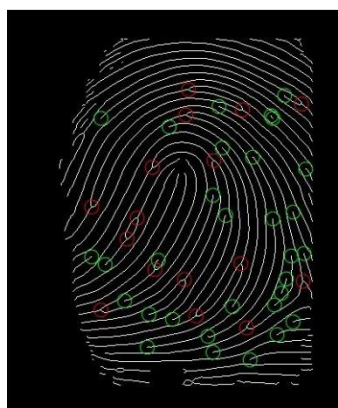
In this exercise, you are going to implement a series of image processing algorithms to extract the features of “Fingerprint.tif”. If interested in, you may further move to perform feature extraction on fingerprints. To extract features, you will need to first improve the quality of the raw image, then use thinning, a morphological algorithm that identifies the skeletons of foreground objects. Only typical features, ridge-ending and bifurcation, are required to be extracted. However, if interested in, you may further move to perform extraction on other features.



Original image



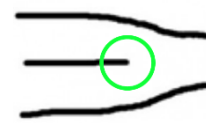
Reconstruct image



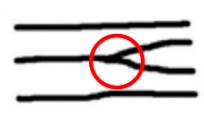
Skeletonize image



Features founded



Ridge-ending



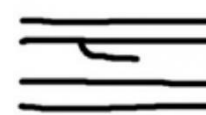
Bifurcation



Dot



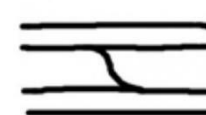
Island



Hook



Eye



Bridge



Trifurcation

F. Properties of Convolution (20 points Bonus)

1. Try to verify the following properties of convolution:

- (a) Commutative $f(x) * g(x) = g(x) * f(x)$
- (b) Associative $f(x) * [g(x) * h(x)] = [f(x) * g(x)] * h(x)$
- (c) Distributive $f(x) * [g(x) + h(x)] = [f(x) * g(x)] + [f(x) * h(x)]$

These properties become important if you think of every operation as a filtering of an image, then you can use a combination of different filters to obtain a particular effect. One example of this would be a sharpening filter that is implemented with an identical filter and a smoothing filter. Remember as well that the convolution is a linear operator, so the combination of convolutional filters is linear as well.

2. Can you think of a non-linear filter? Would it need a special filter or a special operation?